

Correlation filter tracking

Marko Medved

I. INTRODUCTION

In this assignment Note that all measurements were performed on an AMD Ryzen 7 7435HS CPU.

II. EXPERIMENTS

A. Base tracker performance results

Firstly, the simplified correlation filter tracker was implemented and evaluated using the Tracking toolkit-lite and the VOT 2014 dataset. We can see the results in Table I. Note that the used parameters here were: $\sigma = 1.5$ and $\alpha = 0.3$, where σ is the parameter of the Gaussian and α is the update speed parameter. Note that the regularization parameter was set to 2000 (this value is used in all evaluations in this assignment).

Table I

PERFORMANCE OF THE BASIC CORRELATION FILTER TRACKER.

Metric	Value
Average Overlap	0.51
Total Failures	61.0
Average Speed (FPS)	1052.08

B. Testing different parameters

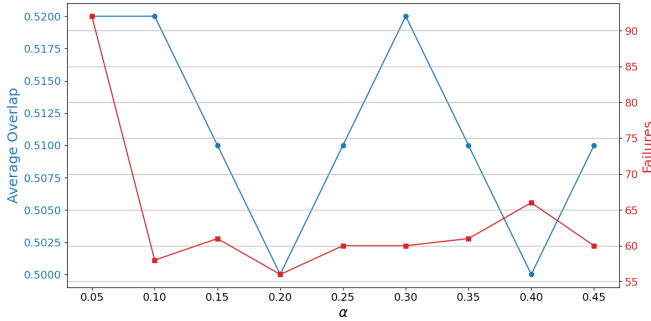


Figure 1. Comparison of the tracker with different update speeds

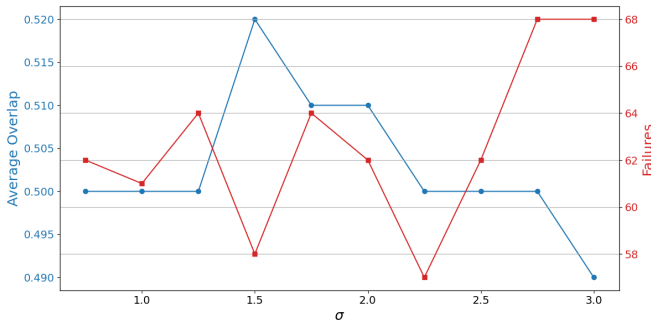


Figure 2. Comparison of the tracker performance with different standard deviations of the Gaussian

C. Testing larger patches

more bg - less failures, worse overlap since the tracker tracks bg

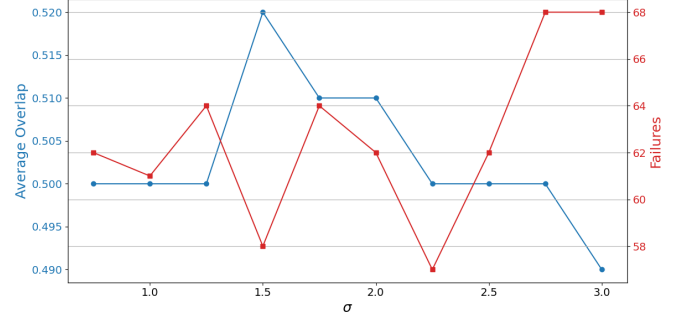


Figure 3. Comparison of the tracker performance with different background enlargement factors

III. CONCLUSION

In this assignment, the mean shift algorithm and tracker were implemented. The analysis identified the problematic test sequences and explored the potential reasons for failure. The results demonstrated that the tracker's performance can be significantly improved by carefully tuning the parameters and incorporating background information into the model.